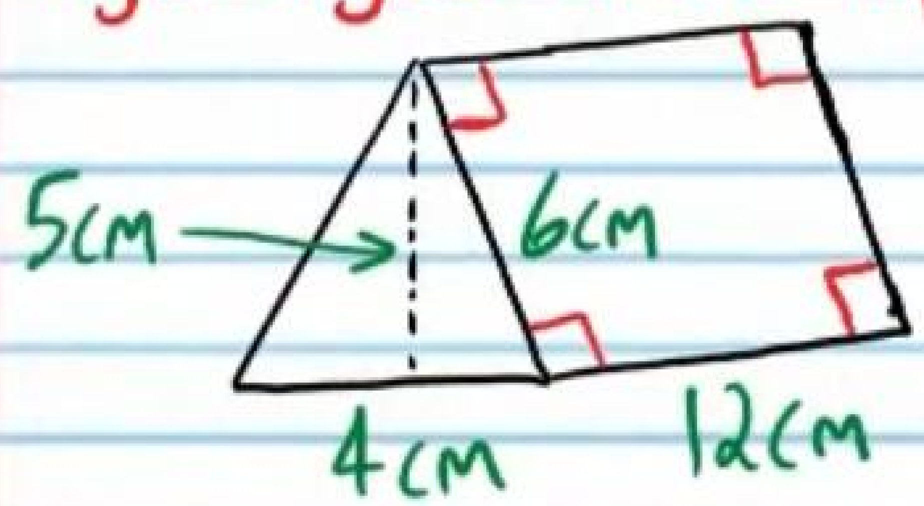
Name
Date
Course

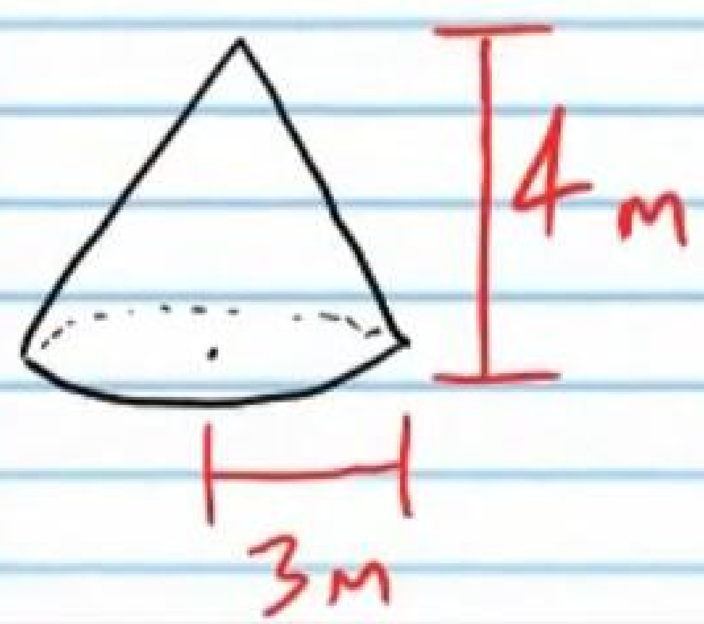
Geometry Test VI

I) Write your name, the date, and the course title (i.e. Geometry).
II) Show all work.
III) If a problem requires algebraic steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
IV) You will need a calculator for some problems on this test. When pi is required, use the π button on your calculator.
V) Each problem is worth four points for a total of 100 points.

① Find the surface area of the prism below. Assume right angles where apparent.

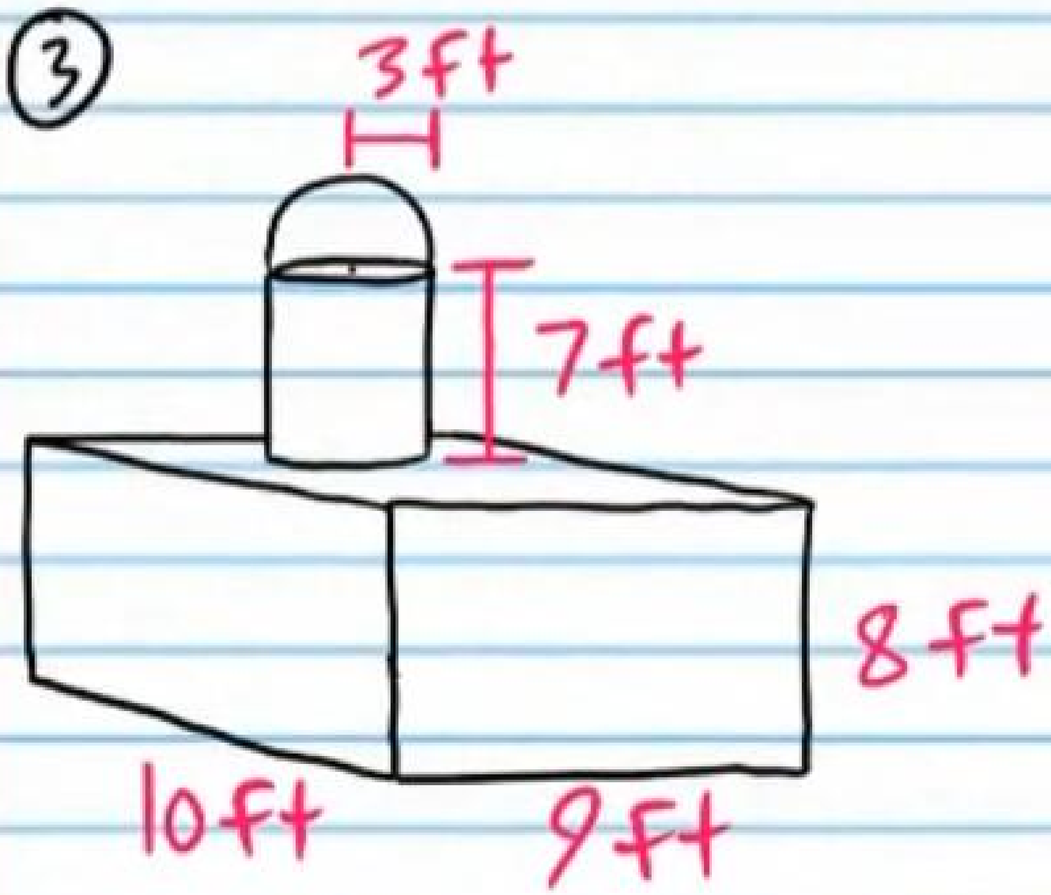


② Find the surface area of the right cone below. Leave your answer as a multiple of pi or round your answer to the nearest tenth.

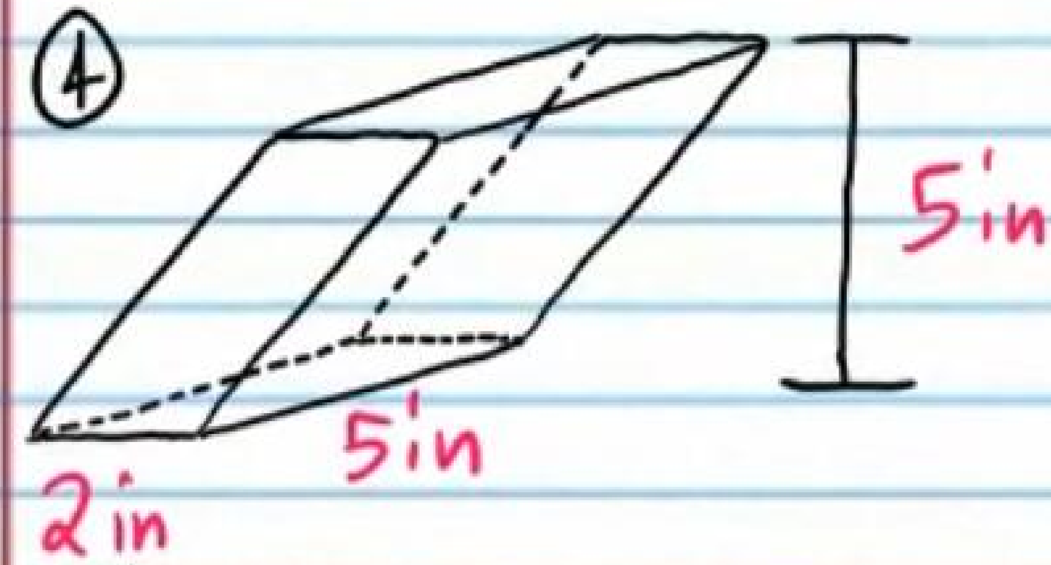


Find the volume of the following figures. Round your answers to the nearest tenth if necessary.

③

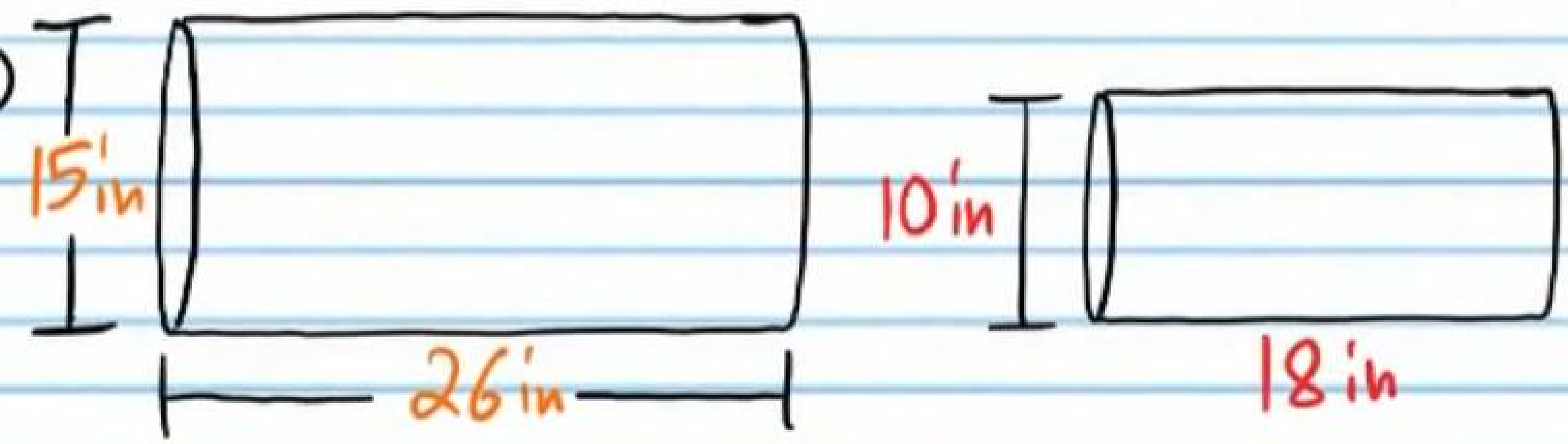


④

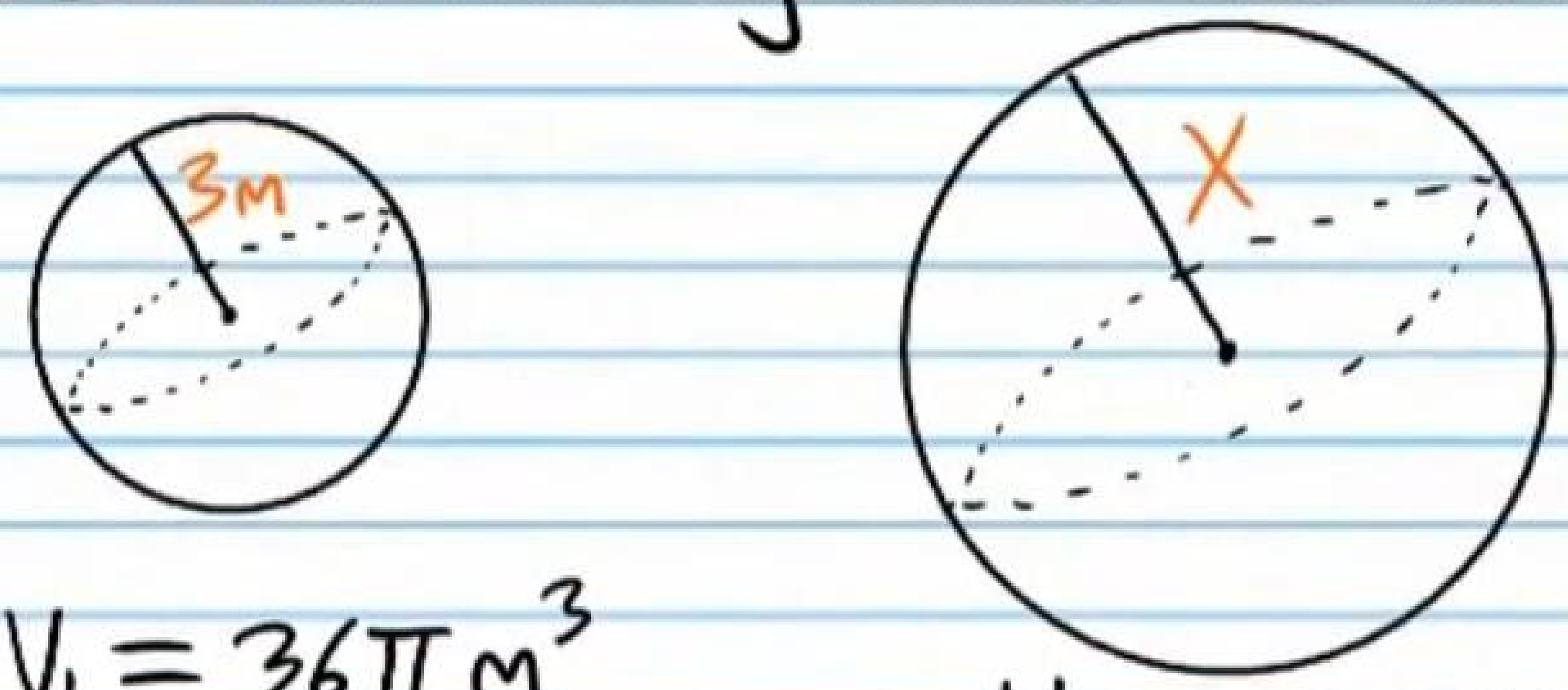


Determine if each pair of figures below contains similar solids. Write the ratios that prove your answers.

⑤



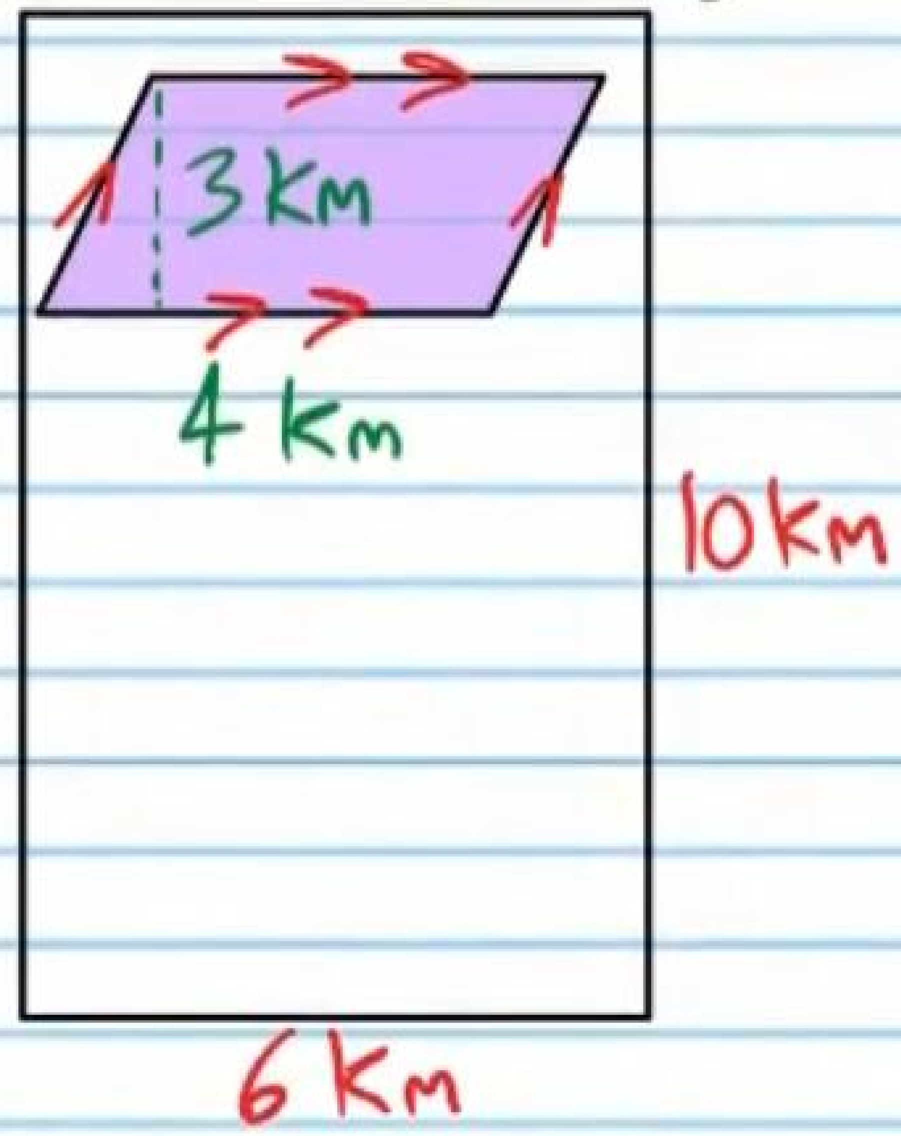
⑥ If the following solids are similar, find X.



$V_1 = 36\pi\ m^3$

$V_2 = 288\pi\ m^3$

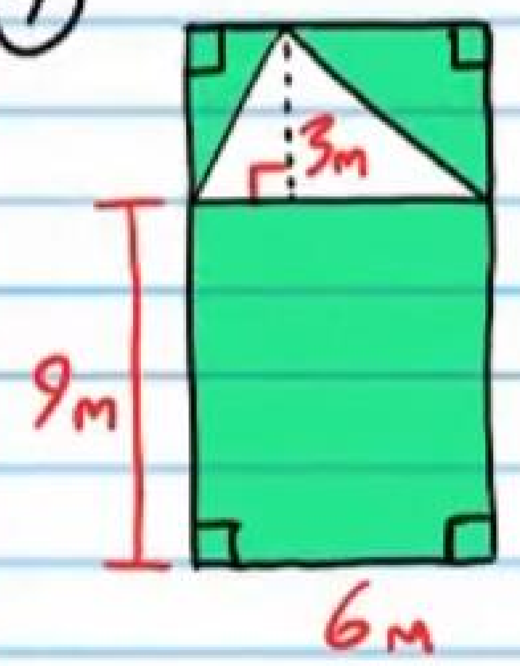
⑦ Find the probability that a randomly chosen point in the figure will lie in the shaded region.



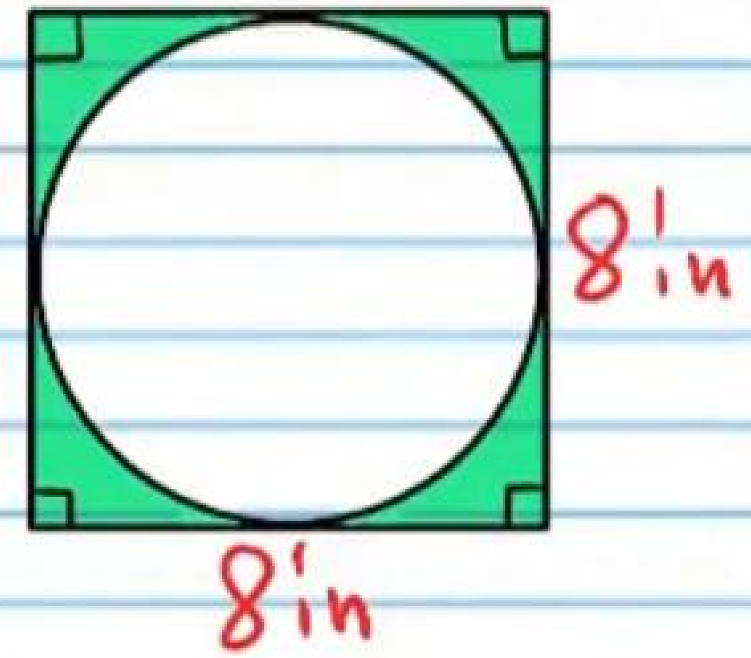
⑧ Write a flowchart proof for the following statement: If $BC=8$ and $AC=10$ in the figure, then $AB=2$.

Find the area of each shaded region below.

⑨

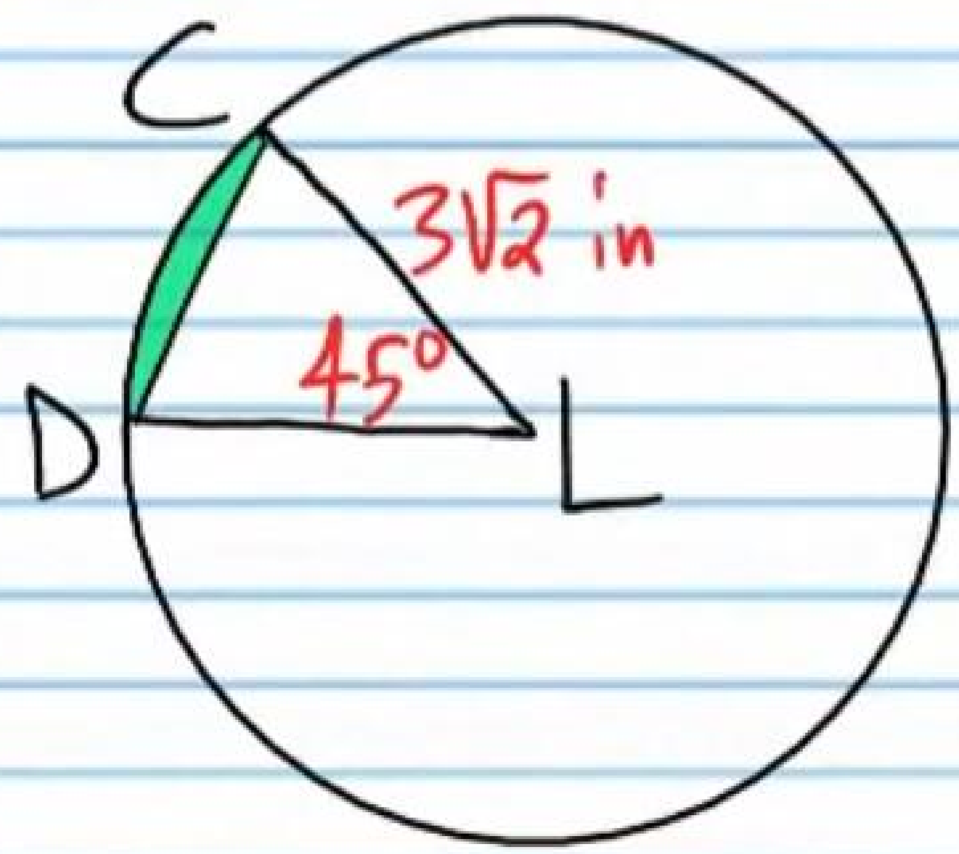


⑩ You may write your answer in terms of π or you may round to the nearest tenth.



Use $\odot L$ below to answer problem #'s 11–13.

⑪ Find the area of sector CLD. Round your answer to the nearest ten thousandth.

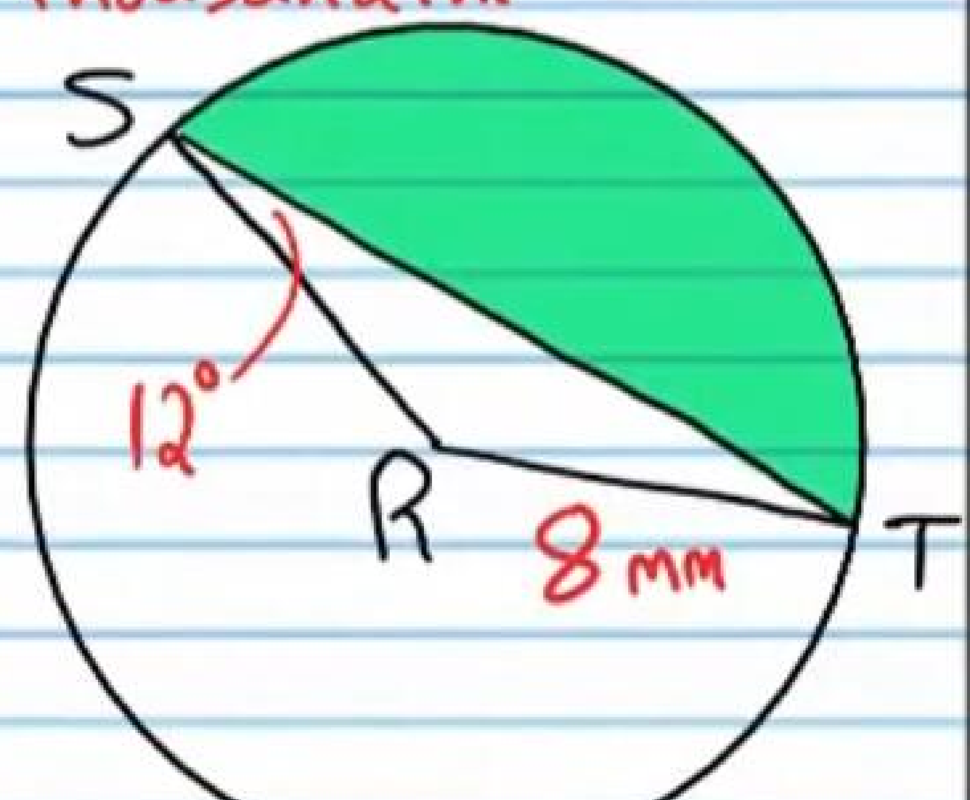


⑫ Find the area of ΔCLD. Round your answer to the nearest ten thousandth or write an exact expression.

⑬ Find the area of the segment (i.e. the shaded region). Round your answer to the nearest tenth.

Use $\odot R$ below to answer problem #'s 14–16.

⑭ Find the area of sector TRS. Round your answer to the nearest ten thousandth.

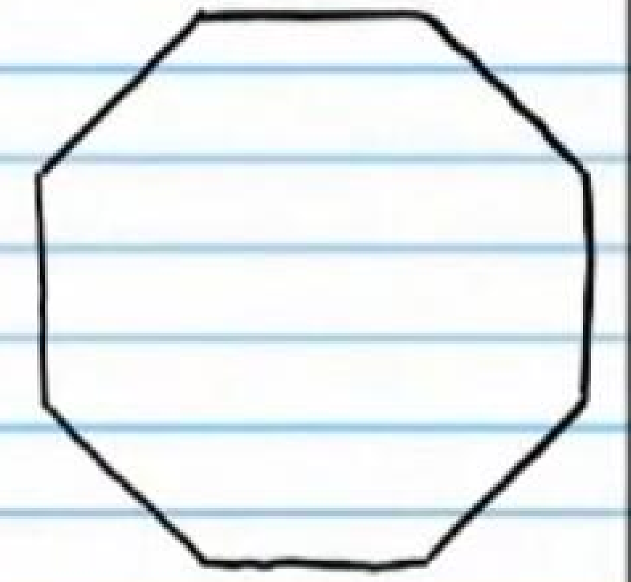


⑮ Find the area of $\triangle TRS$. Round your answer to the nearest ten thousandth.

⑯ Find the area of the segment (i.e. the shaded region). Round your answer to the nearest tenth.

Use the regular polygon below to answer problem #'s 17–18.

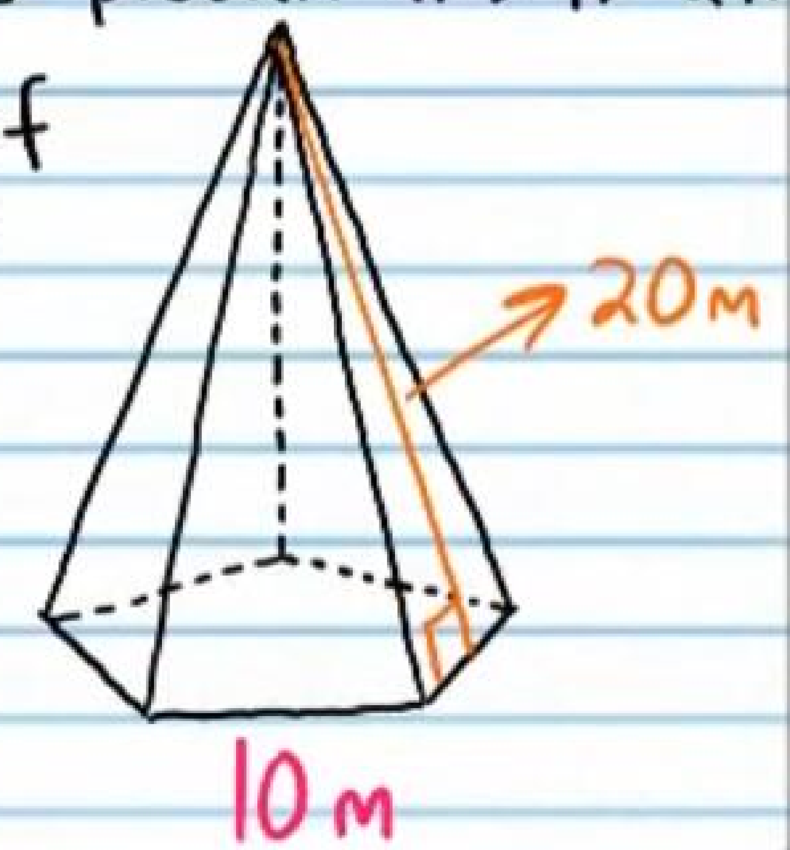
⑰ Find the length of the i) apothem and the ii) side lengths if the radius is 6 cm. Round your answers to the nearest ten thousandth.



⑱ Find the area of the polygon. Round your answer to the nearest tenth.

Use the regular pyramid below to answer problem #'s 19–21.

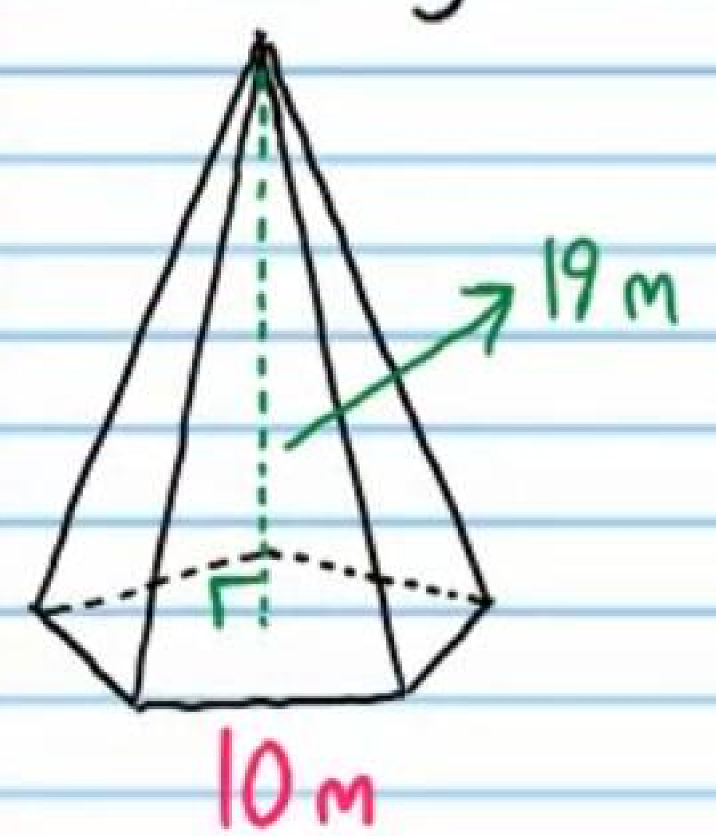
⑲ Find the total area of all of the sides of the pyramid (not including the base).



⑳ Find the area of the base of the pyramid.
Round your answer to the nearest ten thousandth.

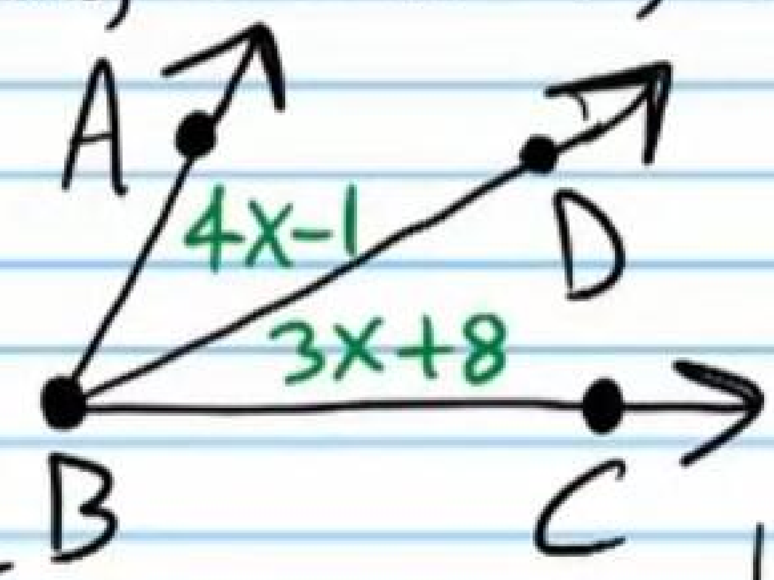
㉑ What is the total surface area of the pyramid?
Round your answer to the nearest tenth.

㉒ Find the volume of the pyramid in problem #'s 19-21 if its height is 19 meters. Round to the nearest tenth.



Use the two-column proof below to do problem #23.

In the figure below, if $x=9$, then $\overrightarrow{BD}$ bisects $\angle ABC$.

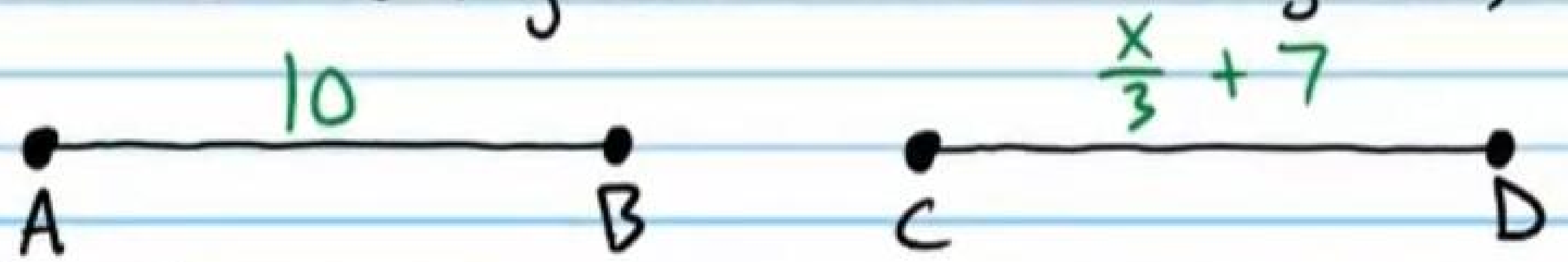


Statement	Reason
I) $x=9$	Given.
II) $m\angle ABD = 4x-1$ & $m\angle DBC = 3x+8$.	Diagram
III) $m\angle ABD = 4(9)-1$ & $m\angle DBC = 3(9)+8$.	Subs. Prop of =.
IV) $m\angle ABD = 35°$ & $m\angle DBC = 35°$	Simplification.
V) $m\angle ABD = m\angle DBC$	Trans. Prop of =.
VI) $\overrightarrow{BD}$ bisects $\angle ABC$	Def. of $\angle$ bisector.

㉓ Convert the two-column proof above to paragraph form.

Use the statement below to do problem #'s 24-25.

If the two segments below are congruent, then $x = 9$.



(24) Write a flowchart proof for the statement above.

(25) Write a paragraph proof for the statement.

Geometry Test VI Answers

1. $\boxed{212\ cm^2}$
2. $\boxed{24\pi\ m^2}$ or $\boxed{75.4\ m^2}$
3. $\boxed{974.5\ ft^3}$
4. $\boxed{50\ in^3}$
5. No. $\frac{15}{10} \neq \frac{26}{18}$

   $\frac{3}{2} \neq \frac{13}{9}$
6. $\boxed{6\ m}$
7. $\boxed{0.2}$ or $\boxed{20\%}$ or $\boxed{1/5}$
8. $BC=8$ — Given.

   $AC=10$ — Given.

   $AC=AB+BC$ — Seg. Add. Post.

   $\downarrow$

   $10=AB+8$ — Subs. Prop of =.

   $\downarrow$

   $2=AB$ — Subt. Prop of =. $\longrightarrow$ $AB=2$ — Symm. Prop of =.

9. $\boxed{63\ m^2}$
10. $\boxed{64-16\pi\ in^2}$ or $\boxed{13.7\ in^2}$
11. $\boxed{7.0686\ in^2}$
12. $\boxed{6.364\ in^2}$
13. $\boxed{0.7\ in^2}$
14. $\boxed{87.1268\ mm^2}$
15. $\boxed{13.0156\ mm^2}$
16. $\boxed{74.1\ mm^2}$
17. i) Apothem: $\boxed{5.5433\ cm}$ ii) Side Length: $\boxed{4.5922\ cm}$
18. $\boxed{101.8\ cm^2}$
19. $\boxed{500\ m^2}$
20. $\boxed{172.0477\ m^2}$
21. $\boxed{672\ m^2}$
22. $\boxed{1{,}089.6\ m^3}$
23. It is given that $x=9$ and the diagram shows that $m\angle ABD = 4x-1$ and $m\angle DBC = 3x+8$. By the Substitution Property of Equality, $m\angle ABD = 4(9)-1$ and $m\angle DBC = 3(9)+8$. If the equations are simplified, they become $m\angle ABD = 35^\circ$ and $m\angle DBC = 35^\circ$ and therefore by the

Transitive Property of Equality, $m\angle DBC = m\angle ABD$. Consequently, $\overrightarrow{BD}$ bisects $\angle ABC$ by the definition of an angle bisector.

(24) $\overline{AB} \cong \overline{CD}$

Given.

$\downarrow$

$AB = CD$

Def. of $\cong$ Segs.

$AB = 10$ & $CD = \frac{x}{3} + 7$

Diagram

$\downarrow$

$10 = \frac{x}{3} + 7$

Subs. Prop of =.

$\downarrow$

$3 = \frac{x}{3}$ $\longrightarrow$ $9 = x$ $\longrightarrow$ $x = 9$

Subt. Prop of =. Mult. Prop of =. Symm. Prop of =.

(25) It is given that $\overline{AB} \cong \overline{CD}$ and the diagram shows that $AB = 10$ and $CD = x/3 + 7$. By the definition of congruent segments, $AB = CD$ and by the Substitution Property of Equality, $10 = x/3 + 7$. Furthermore, by the Subtraction Property of Equality, $3 = x/3$ and by the Multiplication Property of Equality, $9 = x$. Lastly, by the Symmetric Property of Equality, $x = 9$.